

Multiplying and Dividing in Scientific Notation

Question Bank 4.02

Show your work. Circle your final answer.

For questions 1 through 6, multiply or divide. Write each answer in scientific notation.

1. $(2 \times 10^3)(4 \times 10^5)$

2. $\frac{8.8 \times 10^9}{2.2 \times 10^4}$

3. $(6 \times 10^4)(5 \times 10^7)$

4. $\frac{1.5 \times 10^6}{5 \times 10^9}$

5. $(3.5 \times 10^{-6})(4 \times 10^{-3})$

6. $\frac{4.8 \times 10^{-2}}{1.6 \times 10^{-7}}$

Real quantities.

7. Three meteorites have masses 1.1×10^2 , 6.8×10^2 and 8.4×10^{-2} grams.
- How many times as heavy as the lightest one is the heaviest one? Round to the nearest whole number.
 - How many times as heavy as the lightest one is the 1.1×10^2 gram one? Round to the nearest whole number.
8. A data center processes about 4.5×10^6 records per hour.
- How many records does it process in 2.4×10^2 hours?
 - A second center processed 9.0×10^9 records in 1.8×10^3 hours. How many records per hour is that?

Working backwards.

9. Find the missing number in each equation.

a. $(3 \times 10^n)(2 \times 10^4) = 6 \times 10^{11}$

b. $\frac{9.6 \times 10^8}{n \times 10^3} = 1.2 \times 10^5$

c. $(5 \times 10^4)(4 \times 10^6) = \underline{\hspace{2cm}}$

Reading someone else's work.

10. Tyree multiplies $(2.8 \times 10^3)(9.7 \times 10^6)$ and writes 27.16×10^9 . Is his value correct? Is his answer written in scientific notation? Write the finished answer.

11. Which is the value of $\frac{7.5 \times 10^{-4}}{2.5 \times 10^2}$?

- ☐ A. 5×10^{-6}
- ☐ B. 3×10^{-6}
- ☐ C. 3×10^6
- ☐ D. 3×10^{-2}

12. Select ALL of the expressions whose value is 6×10^8 .

- ☐ A. $(1.5 \times 10^4)(4 \times 10^4)$
- ☐ B. $\frac{3 \times 10^{10}}{5 \times 10^1}$
- ☐ C. $(2 \times 10^3)(3 \times 10^5)$
- ☐ D. $\frac{1.2 \times 10^{12}}{2 \times 10^4}$
- ☐ E. $(2 \times 10^3)(3 \times 10^{15})$
- ☐ F. $\frac{1.2 \times 10^{12}}{2 \times 10^3}$